

Chapter 1: Finite reflection groups

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1 Reflections

Exercise 1.1 [Trivial]

The reflections form a single conjugacy class in $\mathcal{D}_m$ when m is odd, but form two classes when m is even.

Exercise 1.2 [Trivial]

Regarding S_n in this way as a subgroup of $O(n, \mathbf{R})$, prove that the transpositions are the sole reflections belonging to S_n .

3 Positive and simple systems

Exercise 3.1 [Trivial]

If Φ has rank 2, prove that W is a dihedral group. [This will be easier to do after Theorem 1.5.]

Exercise 3.2

Find simple systems for the various groups described in 1.1, taking for Φ in each case a convenient set of vectors (not necessarily unit vectors).

Solution. (i) $(I_2(m), m \geq 3)$:

Take

$$\Phi := \{\pm\alpha_j \mid 0 \leq j \leq m-1, \alpha_j := (\cos j\theta, \sin j\theta)\}$$

and we have simple system

$$\Delta := \{\alpha_0, \alpha_{m-1}\}$$

(ii) $(A_{n-1}, n \geq 2)$:

Take

$$\Phi := \{\varepsilon_i - \varepsilon_j \mid i \neq j\}$$

and we have simple system

$$\Delta := \{\varepsilon_i - \varepsilon_{i+1} \mid i \in [1, n-1] \cap \mathbb{Z}\}$$

(iii) $(B_n, n \geq 2)$:

Take

$$\Phi := \{\pm\varepsilon_i, \pm(\varepsilon_i - \varepsilon_j) \mid 1 \leq i < j \leq n\}$$

and we have simple system

$$\Delta := \{\varepsilon_1 - \varepsilon_2, \varepsilon_2 - \varepsilon_3, \dots, \varepsilon_{n-1} - \varepsilon_n, \varepsilon_n\}$$

(iv) $(D_n, n \geq 4)$:

Take

$$\Phi := \{\pm(\varepsilon_i \pm \varepsilon_j) \mid 1 \leq i < j \leq n\}$$

and we have simple system

$$\Delta := \{\varepsilon_1 - \varepsilon_2, \varepsilon_2 - \varepsilon_3, \dots, \varepsilon_{n-1} - \varepsilon_n, \varepsilon_{n-1} + \varepsilon_n\}$$

All of them are easy to examine, and we're done.

5 Generation by simple reflections

Exercise 5.1 [Trivial]

Let Φ be a root system of rank n consisting of unit vectors. If $\Psi \subset \Phi$ is a set of n roots whose mutual angles agree with those between the roots in some simple system, then Ψ must be a simple system.

Proof. (Use **Corollary 1.5**.) □

Exercise 5.2 [Skipped]

Given a simple system Δ , no proper subset of the simple reflections can generate W . [Otherwise find $\alpha \in \Delta$ for which s_α is not needed as a generator of W . Consider $w \in W$ for which $w(-\alpha) \in \Delta$.]

Exercise 5.3 [Skipped]

If $\beta \in \Pi \setminus \Delta$, prove that $\text{ht}(\beta) > 1$.

6 The length function

Exercise 6.1

(a) If $w \in W$, prove that $\text{Det}(w) = (-1)^{n(w)}$. (b) If $w, w' \in W$, prove that $n(ww') \leq n(w) + n(w')$ and $n(ww') \equiv n(w) + n(w') \pmod{2}$.

Proof. (a) Because from **Lemma 1.6**, we know

$$n(w) \equiv l(w) \pmod{2}.$$

$$\text{So } \text{Det } w := (-1)^{l(w)} = (-1)^{n(w)}.$$

(b) (Again use **Lemma 1.6**, trivial.) □

Exercise 6.2 [Skipped]

Taking the simple reflections in S_n to be the transpositions $(i, i+1)$, show that the length of a permutation π is the number of 'inversions': the number of pairs $i < j$ for which $\pi(i) > \pi(j)$.

7 Deletion and Exchange Conditions

Exercise 7.1

In the Exchange Condition, suppose $\ell(w) = r$. Prove that the index i in the conclusion is uniquely determined.

Proof. Suppose $\exists 1 \leq i < j \leq r$, s.t.

$$ws = s_1 \dots \hat{s}_i \dots s_r = s_1 \dots \hat{s}_j \dots s_r.$$

From which we have

$$\hat{s}_i s_{i+1} \dots s_{j-1} s_j = s_i s_{i+1} \dots s_{j-1} \hat{s}_j,$$

i.e.

$$s_i \dots s_j = s_{i+1} \dots s_{j-1}$$

and

$$s = s_1 \dots \hat{s}_i \dots \hat{s}_j \dots s_r$$

contradicting the minimality of r . □

Exercise 7.2 [Skipped]

Formulate a 'left-handed' version of the Exchange Condition, under the hypothesis $\ell(sw) < \ell(w)$.

Solution. Let $w = s_1 \cdots s_r$ (not necessarily reduced), where each s_i is a simple reflection. If $\ell(sw) < \ell(w)$ for some simple reflection s , then there exists an index i for which $sw = s_1 \cdots \hat{s}_i \cdots s_r$ (and thus $w = ss_1 \cdots \hat{s}_i \cdots s_r$, with a factor s exchanged for a factor s_i). In particular, w has a reduced expression beginning with s if and only if $\ell(sw) < \ell(w)$.

(I don't wanna prove it. Seemed to be trivial.)

8 Simple transitivity and the longest element

Exercise 8.1

What is w_0 in the case of S_n , relative to the simple system $\varepsilon_1 - \varepsilon_2, \dots, \varepsilon_{n-1} - \varepsilon_n$?

Solution. In S_n , $\Pi = \{s_{\varepsilon_i - \varepsilon_j} \mid i < j\}$. To send Π to $-\Pi$, the required w_0 is

$$w_0 = \begin{pmatrix} 1 & 2 & \dots & n \\ n & n-1 & \dots & 1 \end{pmatrix}$$

And we know

$$n(w_0) = \frac{n(n-1)}{2}.$$

Exercise 8.2 [Skipped]

In any reduced expression for w_0 , every simple reflection must occur at least once.

10 Parabolic subgroups and minimal coset representatives

Exercise 10.1 [Skipped]

Is there a result analogous to (c) describing minimal representatives of the double cosets $W_I w W_J$ ($I, J \subset S$)?

Exercise 10.2 [Skipped]

When $W = S_n$, prove that each parabolic subgroup of W is isomorphic to a direct product of symmetric groups.

Exercise 10.3 [Skipped]

Given $s \neq s'$ in S , set $v := ss'ss' \cdots$ (m factors, where m is the order of ss' in W), so also $v = s'ss's \cdots$ (m factors), and $v^2 = 1$. If $w \in W$ satisfies $\ell(ws) < \ell(w)$ and $\ell(ws') < \ell(w)$, prove that $\ell(wv) = \ell(w) - m$. [Consider W_I , $I = \{s, s'\}$. Since $w = (wv)v$ with $v \in W_I$, it suffices to show that $wv \in W^I$. Look at the action on the roots corresponding to s, s' .]

11 Poincare polynomials

Exercise 11.1

When $W = S_3$, use the formula in the proposition to compute $W(t)$ inductively, starting with the fact that $W(t) = 1 + t$ for a group of rank 1. Do the same for $W = \mathcal{D}_m$ in general.

Proof. Notice from

$$\sum_{I \subseteq S} (-1)^I \frac{W(t)}{W_I(t)} = t^N$$

we get

$$\sum_{I \subseteq S} \frac{(-1)^I}{W_I(t)} = \frac{t^N}{W(t)}.$$

Breaking up left hand side, we obtain

$$\frac{(-1)^S}{W(t)} + \sum_{I \subsetneq S} \frac{(-1)^I}{W_I(t)} = \frac{t^N}{W(t)},$$

i.e.

$$\frac{t^N - (-1)^S}{W(t)} = \sum_{I \subseteq S} \frac{(-1)^I}{W_I(t)}$$

(Rest of the proof is pure repetitive work, skipped.)

□

Exercise 11.2 [Skipped]

The identity just obtained permits an inductive calculation of $|W|$ when $|S|$ is odd. Suppose for example that $|S| = 3$, and that the dihedral subgroups W_I are of respective orders 4, 6, 10. What is $|W|$?

12 Fundamental domains

Exercise 12.1

Show how the theorem can be used to solve the Word Problem for W : given a product of simple reflections, decide whether or not it equals 1 in W .

Solution. Pick out an element in the corresponding C of the given W and see if it's fixed by such product of simple reflections: if so, then the product is equal to 1.